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EnvGeo-Seawater: An Interactive Platform for Exploring Seawater Isotope and Hydrographic Data

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Software

- Review
- Repository
- Archive

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Summary

EnvGeo-Seawater is a web-based interactive visualization platform for exploring marine geochemical and hydrographic datasets, including stable water isotopes (δ^{18}O and δD), salinity, temperature, and depth.

The platform integrates approximately 50,000 seawater isotope records from major global datasets, including the NASA GISS database (Schmidt et al., 1999) and the CoralHydro2k seawater isotope database (Atwood et al., 2026), together with cited regional reference data and the internally consistent **EnvGeo Dataset [ECS–Japan Sea]** core collection (primary reference: Kodama et al. 2024) (Kodama et al., 2024).

EnvGeo-Seawater enables simultaneous exploration of spatial distributions, cross-variable relationships, and vertical structures through an integrated interface. By combining mapping, depth profiles, temperature–salinity diagrams, regression analysis, and multi-dimensional (3D/4D) plots, it supports rapid exploratory analysis and documented comparison of heterogeneous seawater datasets.

Statement of Need

Seawater isotope measurements (e.g., δ^{18}O, δD, and d-excess) are widely used in oceanography and paleoclimate research to investigate ocean circulation, freshwater fluxes, and climate processes. However, despite the availability of large public datasets, integrated and interactive analysis across multiple variables and datasets remains limited.

Although public datasets provide essential archival access, researchers often still assemble custom and fragmented workflows to explore relationships among isotopic and hydrographic variables across separately curated sources.

EnvGeo-Seawater addresses this gap by providing a unified, interactive environment that integrates heterogeneous global datasets, cited regional reference data, and an internally consistent Japan-region core collection. Source-specific citations and provenance remain visible so that comparisons can be interpreted in light of their respective sampling and analytical contexts.

Integrating independently curated reference datasets also creates a practical quality-assurance challenge: records inherited from a common original source, or records affected by rounding and revised metadata, can resemble duplicates. EnvGeo-Seawater therefore provides a provenance-aware, read-only cross-dataset overlap screen. It uses explicit user-visible criteria for matching sampling time and for comparing coordinates, depth, salinity, and δ^{18}O; it separates strong candidates from cases requiring review and exports the matching evidence for inspection. Source records are preserved and candidate status is not treated as confirmation of duplication. The shared filter sidebar retains all records by default and offers reversible, opt-in sensitivity

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screens: a deterministic one-to-one subset whose recorded differences are compatible with inferred rounding precision, and a broader Strong-candidate screen that is explicitly not a confirmed de-duplication result. The default thresholds are user-visible and adjustable; they are screening criteria rather than universal measurement-error thresholds. This makes cross-dataset integration more transparent while retaining the source-level context needed for scientific interpretation.

The locally installable application also supports a reproducible workflow that does not depend on the continued availability of a hosted web service: bundled-data loading, filtering, analysis, and figure generation can be run locally after installation. Online basemap tiles and external source links remain network-dependent features and are not represented as offline data.

A key contribution is the integration of the regionally curated EnvGeo Dataset [ECS–Japan Sea] collection, currently represented by Kodama et al. (2024), which provides a consistent analytical baseline for exploratory comparison across its sampled locations and periods.

State of the field

Oceanographic and geochemical datasets, particularly those including seawater stable isotopes (e.g., δ^{18}O, δD), are increasingly available through global and regional databases such as the NASA GISS seawater isotope database and CoralHydro2k. However, these datasets are often distributed across heterogeneous formats and lack integrated tools for interactive exploration.

Existing oceanographic visualization tools, such as Ocean Data View (ODV) (Schlitzer, 2002), provide powerful capabilities for analyzing hydrographic data. EnvGeo-Seawater complements rather than replaces such general-purpose tools: it provides a lightweight, local, seawater-isotope-specific workflow across bundled cited sources and session-only user-supplied data, while retaining source-specific provenance and citations, supporting an overlap-candidate audit, and providing figure-oriented interactive exploration. This specialization supports comparison of isotope and hydrographic variables across heterogeneous, cited seawater datasets without first assembling a workflow from general-purpose components.

Software design

EnvGeo-Seawater is organized as a modular Python application with shared modules for asset resolution, data loading, filtering, and visualization, plus Streamlit page scripts for the interactive interface. These modules are reused within the application, while the interactive interface is the primary analysis route.

The design supports the addition of new, source-documented datasets and future maintenance without changing the existing page-oriented workflow. Installed console commands start the application or its local diagnostic tool.

The interactive web interface is implemented using Streamlit, which provides an accessible platform for exploratory analysis. The application supports multiple visualization types, including map-based exploration, temperature–salinity diagrams, depth profiles, and regression analyses.

A figure-supported, page-by-page user guide is available in English and Japanese at https://envgeo.github.io/seawater_map/.

Oceanographic colour scales follow the cmocean design guidance (Thyng et al., 2016). T–S diagrams include approximate σ_0 reference contours calculated with the Gibbs SeaWater (GSW) implementation of TEOS-10 (McDougall & Barker, 2011), using Practical Salinity and in-situ temperature as display-oriented proxies. They are visual reference contours rather than fully converted TEOS-10 density values for individual observations. The Vertical Section Visualizer can use a project-derived, downsampled GEBCO 2025 Grid for bathymetric context (GEBCO Compilation Group, 2025).

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Research impact

Before the software received an archival DOI, EnvGeo-Seawater was used in the author's and collaborators' research workflows to select, explore, and visualize subsets of the regional seawater isotope dataset reported by Kodama et al. (2024). These workflows supported isotope-based interpretation in studies of Japanese sardine migration (Aono et al., 2024), sardine juvenile habitat selection (Sakamoto et al., 2024), anguillid eel larval experienced-temperature reconstruction (Kuroki et al., 2025), and early-life Japanese jack mackerel movement (Sakamoto et al., 2026). The resulting publications cite the underlying Kodama et al. (2024) dataset paper rather than EnvGeo-Seawater; they are evidence of workflow use, not direct software citations. The platform has also been used in four scientific conference presentations in Japan. For example, a 2024 presentation introducing visualisation of seawater stable-isotope data around Japan and the online interactive database is publicly archived (Ishimura, 2024).

Future directions

The data model is designed to accommodate additional datasets after their sources, provenance, and redistribution status have been recorded. Reusable visualization, asset-resolution, and distribution components may also support future related EnvGeo applications. These are future directions and are not part of the v1.3.5 stable release scope.

Capabilities

The platform provides the following capabilities:

- Interactive spatial mapping with adaptive zoom
- Shared interactive filtering by geographic extent, sampling year and month, depth, dataset, and citation
- Depth profile visualization with gap-aware plotting for discrete sampling data
- Temperature–salinity (T–S) diagrams with approximate σ_0 reference contours
- Cross-variable analysis (e.g., salinity–δ^{18}O relationships with regression)
- Consistent calculation and visualization of d-excess from paired δ^{18}O and δD measurements, where available, across spatial, profile, and multi-dimensional views
- Multi-dimensional visualization (3D/4D exploration of spatial–temporal structures)
- Integration of global datasets (~50,000 records), cited regional reference data, and the EnvGeo Dataset [ECS–Japan Sea] core collection (primary reference: Kodama et al. 2024)
- Provenance-aware dataset and citation selection, transparent quality/exclusion reporting, and overlap screening with explicit criteria, strong/review candidate classes, downloadable audit evidence, and reversible display sensitivity screens
- Session-only user-data upload, quality checking, and direct comparison with reference datasets
- Optional local-coastline map fallback for offline or degraded-network use
- Export of publication-quality figures

Implementation

The software is implemented in Python using Streamlit (Streamlit Inc., 2024) for the web interface, Plotly (Plotly Technologies Inc., 2024) for interactive visualization, and Matplotlib

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(Hunter, 2007) for high-quality figure generation. The codebase is modular and designed to support extension to additional datasets and visualization methods.

The bundled data and visual assets occupy approximately 30 MB when unpacked. Local execution requires installation of the declared Python dependencies. The package declares Python >=3.10; wheel builds and isolated installs are verified in continuous integration on Python 3.10 and 3.12, with Python 3.12 recommended for local installation. The v1.3.5 release is published on PyPI: users can install it with `python -m pip install envgeo-seawater` and start the local application with `envgeo-seawater`.

Availability

The stable public source repository is https://github.com/envgeo/seawater_map. Version 1.3.5 was released on 17 October 2026 (GitHub Release: https://github.com/envgeo/seawater_map/releases/ta and is distributed through PyPI at https://pypi.org/project/envgeo-seawater/1.3.5/. The version-specific archival record is https://doi.org/10.5281/zenodo.23428743; the all-versions concept DOI is https://doi.org/10.5281/zenodo.23117783.

Example Use Case

EnvGeo-Seawater supports exploratory analysis in research and education settings across spatial and temporal scales, including visualization of global δ^{18}O distributions, salinity–isotope relationships, and vertical structures.

The platform also enables direct comparison between user-provided datasets and curated reference datasets within a unified analytical framework.

A particularly important use case is rapid data checking aboard research vessels, where satellite connectivity may be limited or unavailable. After local installation, researchers can use bundled reference data and local measurements to inspect data quality, sampling positions, depth profiles, temperature–salinity structure, and isotope–hydrographic relationships. Detecting outliers, coordinate errors, missing values, or unexpected profiles while still at sea can inform remeasurement, additional sampling, and adjustments to the remaining observation plan.

AI Usage Disclosure

Development of this software, from version 1.3 onward, has made substantial use of AI coding assistants for code review, implementation drafting, refactoring, test design and authoring, bug investigation, and documentation. The tools used were OpenAI Codex and Anthropic Claude Code (principal models: GPT-5.6 Terra and Claude Sonnet 4.6, respectively). Portions of language editing in earlier drafts of this manuscript were also assisted by ChatGPT (OpenAI).

All AI-assisted output was reviewed, edited, and verified by the corresponding author before being adopted into the codebase, tests, documentation, or this manuscript. The author performed all scientific design, data interpretation, and validation decisions, and takes full responsibility for the accuracy, originality, licensing, and ethical and legal compliance of the software and this manuscript. AI tools are not authors or co-developers of this work.

Conflict of Interest

The author declares no conflicts of interest.

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